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CORRECT ANSWER SOLVED BY HADI
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Best of luck!



Question No : 1 of 26

Marks: 1 (Budgeted Time 1 Min)

The solutions $y_1 = c_1$, $y_2 = c_2 \cos x$, $y_3 = c_3 \sin x$ of a Differential Equation are

Answer (Please select your correct option)

☐ Linearly independent



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☐ Linearly dependent

☐ both of the other options

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Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min)

Wronskian of the functions $y_1(x) = 1$, $y_2(x) = \cos x$, $y_3(x) = \sin x$ is

Answer (Please select your correct option)

☐ 0



☐ 1

☐ 2

☐ 3

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Question No : 3 of 26

Marks: 1 (Budgeted Time 1 Min)

The value of amplitude in the solution $X=7\sin(3t+2.45)$ is

Answer (Please select your correct option)

- ☐ 7
- ☐ 3
- ☐ 2.45
- ☐ 1
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Question No : 4 of 26

Marks: 1 (Budgeted Time 1 Min)

The order of the differential equation $\left(\frac{du}{dt}\right)^{\frac{1}{3}} = \frac{1}{u}$ is

Answer (Please select your correct option)

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
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Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min)

The solution of the differential equation $\frac{dy}{dx} = \tan x$ is

Answer (Please select your correct option)

☐ $y = \ln |\sin x| + c$

☐ $y = \ln |\sec x| + c$

☐ $y = \ln |\operatorname{cosec} x| + c$

☐ $y = \ln |\cos x| + c$



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Question No : 6 of 26

Marks: 1 (Budgeted Time 1 Min)

A differential equation $M(x, y) dx + N(x, y) dy = 0$ is said to be an exact if -----.

Answer (Please select your correct option)

☐ $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$

☐ $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

☐ $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

☐ $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$



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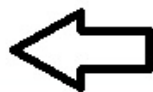
Question No : 7 of 26

Marks: 1 (Budgeted Time 1 Min)

The differential equation $(x^2 - 2x + 2y^2) dx + 2xy dy = 0$ is a/an -----differential equation.

Answer (Please select your correct option)

- ☐ Exact
- ☐ Non-exact
- ☐ Homogenous
- ☐ Separable



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Question No : 8 of 26

Marks: 1 (Budgeted Time 1 Min)

For a 1st order linear differential equation $\frac{dy}{dt} - \frac{1}{t}y = 2t^2$, the integrating factor will be -----.

Answer (Please select your correct option)

☐ $\mu = -\frac{1}{t}$

☒ $\mu = \frac{1}{t}$



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☐ $\mu = t$

☐ $\mu = 2t$

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Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min)

A differential equation $\frac{dy}{dx} + p(x)y = q(x)y^n$ for $n \neq 0, 1$ is called a/an ----.

Answer (Please select your correct option)

☐ Non-Exact differential equation

☐ Linear differential equation

☐ Bernoulli equation

☐ Exact differential equation



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Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min)

In order to change the Bernoulli Equation

$$\frac{dy}{dx} + p(x)y = q(x)y^x$$

into linear differential equation, we choose ----.

Answer (Please select your correct option)

☐ $v = y^{x-1}$

☒ $v = y^{1-x}$



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☐ $v = y^x$

☐ $v = y'$

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Question No : 11 of 26

Marks: 1 (Budgeted Time 1 Min)

To convert the non-homogeneous differential equation $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$ into separable form, if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$, then we use the substitution-----.

Answer (Please select your correct option)

☐ $x = X + h$

☐ $y = Y + h$

☐ $z = a_1x + b_1y$

☐ $z = Y + X$



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Question No : 12 of 26

Marks: 1 (Budgeted Time 1 Min)

In exponential model for the population growth $P(t) = P_0 e^{kt}$ If $k > 0$, then $\lim_{t \rightarrow \infty} P(t) = \dots$

Answer (Please select your correct option)

- ☐ ∞  correct answer solve by hadi
- ☐ $-\infty$
- ☐ 0
- ☐ 1

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Question No : 13 of 26

Marks: 1 (Budgeted Time 1 Min)

The constant solutions of the logistic equation $\frac{dP}{dt} = P(a - bP)$ are-----.

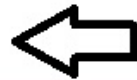
Answer (Please select your correct option)

☐ P=0, P= b/a

☐ P=0, P=a/b

☐ P=a/b, P=b/a

☐ P=0, P=0



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Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min)

The orthogonal trajectory to the family of curves $3x + 4y = c$ is-----.

Answer (Please select your correct option)

- ☐ $y = 2x$
- ☐ $y - x = c$
- ☐ $4x - 3y = c$
- ☐ $4x + 3y = c$



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Question No : 15 of 26

Marks: 1 (Budgeted Time 1 Min)

If the tangent lines of two curves are perpendicular at their point of intersection then both the curves are-----

Answer (Please select your correct option)

☐ Non-intersecting curves

☐ Parallel curves

☐ Orthogonal curves

☐ Intersecting curves



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Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min)

If initial amount of a radioactive isotope is 100g. What will be the amount at the end of 30 days such that $K=0.043$?

Answer (Please select your correct option)

☐ 371.415

☐ 380.560

☐ 363.279

☐ 360.351



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Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)

If $y = 1 + 6x^2 - 7x^3$, then which of the following is true for it ?

Answer (Please select your correct option)

☐ Its annihilator operator is D .

☐ Its annihilator operator is D^2 .

☐ Its annihilator operator is D^3 .

☐ Its annihilator operator is D^4 .



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Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min)

If $y = e^{-6x}$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- ☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)^2$.
- ☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)^3$.
- ☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)$.
- ☐ The annihilator operator of $y = e^{-6x}$ is $(D + 6)$.



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If $y_1 = x$ and $y_2 = xe^x$ are the first and second solution of $x^2 \frac{d^2 y}{dx^2} - x(x+2) \frac{dy}{dx} + (x+2)y = 0$ on $(0, \infty)$, then which of the following is the most accurate option?

Answer (Please select your correct option)

☐ $y_1 = x$ and $y_2 = xe^x$ may or may not be linearly dependent.

☐ $y_1 = x$ and $y_2 = xe^x$ must be linearly independent.

☐ $y_1 = x$ and $y_2 = xe^x$ may or may not be linearly independent.

☐ $y_1 = x$ and $y_2 = xe^x$ must be linearly dependent.



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If $y_2 = x^3$ is the second solution of $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = 0$, then which of the following is the most accurate option?

Answer (Please select your correct option)

- ☐ The first solution say y_1 must be non-trivial on the indicated interval.
- ☐ The first solution say y_1 can be trivial on the indicated interval.
- ☐ The first solution say y_1 must be discontinuous on the indicated interval.
- ☐ The first solution say y_1 may or may not be continuous on the indicated interval.



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Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min)

Find the order of the homogenous equation obtained from the following non-homogenous differential equation

$$y'' + 4y' + 3y = 4x^2 + 5?$$

Answer (Please [click here](#) to Add Answer)



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Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

If half-time life (T) of a radioactive isotope is 5 days, then find k.

Answer (Please [click here](#) to Add Answer)



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$$\frac{dy}{dx} - 5y = 0$$

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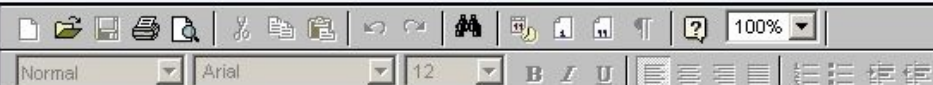
Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)

If $y_1(x) = e^x$ is the first solution of $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} - 3y = 0$, then construct its second solution using $y_2(x) = y_1(x) \int \frac{e^{-\int P(x) dx}}{y_1^2(x)} dx$.

Note: where $P(x)$ is a coefficient of the first derivative involving in the above given differential equation.

Answer (Please [click here](#) to Add Answer)



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Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

Find period of vibration of the solution of a simple harmonic motion?

Answer (Please [click here](#) to Add Answer)



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Using Newton's Law of Cooling with initial condition $T(0)=T_0$ and if temperatures at time t_1 and t_2 are known then show that

$$\frac{T(t_1) - T_m}{T(t_2) - T_m} = e^{k(t_1 - t_2)}$$

Answer (Please [click here](#) to Add Answer)



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